Ⅲ. The foundamental group $I_\cdot = [0,1]$

1. Foundamental Groupoid.

Def Ⅲ.1.1. let $f, g : I \to X$ be paths with $f(1) = g(0)$. Define $f * g$: $I \to X$. where $(f*g)(t) = \begin{cases} f(2t) & 0 \le t \le \frac{1}{2} \\ g(2t-1) & \frac{1}{2} \le t \le 1 \end{cases}$.

Def Ⅲ.1.2. Let $A \subseteq X$. and let $f_0, f_1 : X \to Y$ be continous with $f_0|_A = f_1|_A$

We write $f_0 \cong f_1$ rel A. iff $\exists$ continous $T : X \times I \to Y$ with $T : f_0 \cong f_1$ and $F(a,t) = f_0(a) = f_1(a)$ $\forall a \in A$ $t \in I$.

Rmk. The above homotopy F is called a relative homotopy. It's easy to check that homotopy rel A is an equivalence relation on the set of continuous map: $X \to Y$

Def Ⅲ.1.3. Let $\dot{I} = \{0,1\}$ be the boundary of I in $\mathbb{R}$. The equivalence class of $f : \dot{I} \to X$ $\overset{\text{rel } \dot{I}}{\wedge}$ is called the path class of f. denoted by $[f]$.

Thm Ⅲ.1.4. Assume that f_0, f_1, g_0, g_1 are paths in X. with $f_0 \cong f_1$ rel $\dot{I}$. $g_0 \cong g_1$ rel $\dot{I}$.

If $f_0(1) = g_0(1) = f_1(0) = g_1(0)$. then $(f_0 * g_0) \cong (f_1 * g_1)$ rel $\dot{I}$

